□ Lab Assign 1
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```
from collections import deque

class Node:
    def __init__(self, val):
        self.val = val
        self.left = None
        self.right = None

# BFS
def bfs(root):
    q = deque([root])
    while q:
        node = q.popleft()
        print(node.val, end=" ")
        if node.left:
            q.append(node.left)
        if node.right:
            q.append(node.right)

# DFS (Preorder)
def dfs(root):
    if root:
        print(root.val, end=" ")
        dfs(root.left)
        dfs(root.right)

# Build Tree
root = Node('A')
root.left = Node('B')
root.right = Node('C')
root.left.left = Node('D')
root.left.right = Node('E')
root.right.right = Node('F')

print("BFS:", end=" ")
bfs(root)

print("\nDFS:", end=" ")
dfs(root)
```

```
BFS: A B C D E F
DFS: A B D E C F
```

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['A', 'D', 'E'],
    'C': ['A', 'E'],
    'D': ['B'],
    'E': ['B', 'C']
}

# BFS
def bfs(start):
    visited = set()
    q = deque([start])
    while q:
        node = q.popleft()
        if node not in visited:
            print(node, end=" ")
            visited.add(node)
            q.extend(graph[node])

# DFS
def dfs(node, visited=set()):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)
        for neighbor in graph[node]:
            dfs(neighbor, visited)
```

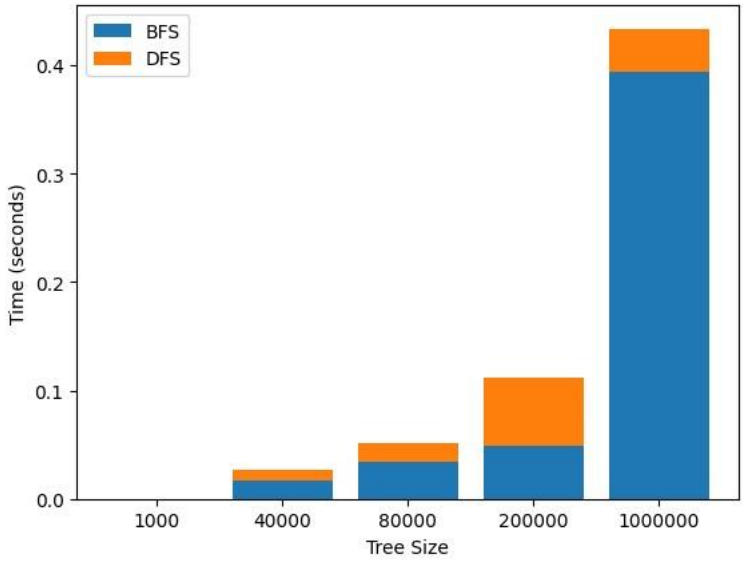
```
print("BFS:", end=" ")
bfs('A')

print("\nDFS:", end=" ")
dfs('A')
```

```
BFS: A B C D E
DFS: A B D E C
```

```
import random
import time
from collections import deque
import matplotlib.pyplot as plt

class Node:
    def __init__(self, val):
        self.val = val
        self.left = None
        self.right = None

def insert(root, val):
    if not root:
        return Node(val)
    if val < root.val:
        root.left = insert(root.left, val)
    else:
        root.right = insert(root.right, val)
    return root

def build_tree(arr):
    root = None
    for num in arr:
        root = insert(root, num)
    return root

def bfs(root, target):
    q = deque([root])
    while q:
        node = q.popleft()
        if node.val == target:
            return True
        if node.left:
            q.append(node.left)
        if node.right:
            q.append(node.right)

def dfs(root, target):
    if not root:
        return False
    if root.val == target:
        return True
    return dfs(root.left, target) or dfs(root.right, target)

sizes = [1000, 40000, 80000, 200000, 1000000]

bfs_times = []
dfs_times = []

for size in sizes:
    data = random.sample(range(size*10), size)
    root = build_tree(data)
    target = data[len(data) - 220]

    start = time.time()
    bfs(root, target)
    bfs_times.append(time.time() - start)

    start = time.time()
    dfs(root, target)
    dfs_times.append(time.time() - start)

# Table
print("\nTree Size | BFS Time | DFS Time")
for i in range(len(sizes)):
    print(f"{sizes[i]} | {bfs_times[i]:.6f} | {dfs_times[i]:.6f}")
```

```
# Graph
plt.bar([str(s) for s in sizes], bfs_times, label='BFS')
plt.bar([str(s) for s in sizes], dfs_times, bottom=bfs_times, label='DFS')
plt.xlabel("Tree Size")
plt.ylabel("Time (seconds)")
plt.legend()
plt.show()
```

```
Tree Size | BFS Time | DFS Time
1000 | 0.000096 | 0.000144
40000 | 0.017478 | 0.009728
80000 | 0.034718 | 0.016908
200000 | 0.048322 | 0.063850
1000000 | 0.394085 | 0.039445
```




```
from collections import deque

def bfs_shortest_path(graph, start, goal):
    visited = set()
    queue = deque([[start]])   # store path

    while queue:
        path = queue.popleft()
        city = path[-1]

        if city == goal:
            return path

        if city not in visited:
            visited.add(city)

            for neighbor in graph[city]:
                new_path = list(path)
                new_path.append(neighbor)
                queue.append(new_path)

    return None

# Run
path = bfs_shortest_path(graph, "A", "D")
print("Shortest Path:", " → ".join(path))
```

```
Shortest Path: A → B → D
```

References:

You Tube: https://www.youtube.com/watch?v=WDlsBx0YmOA&t=2s